Certified AC Robust Secure Sets with GENCO in the Loop

Supplement: complete formulation and proofs

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Abstract

This supplement states every equation used by the code of the entry (DC model of the robust secure set, bilevel and KKT form, the big-M baseline, the two-phase Benders decomposition, the AC model and correction loop, GENCO's corrective-layer trajectory, physics-trajectory recalibration with split-conformal screening, the gradient adversary and the battery extension) and gives complete proofs of all results: the meaning of the RSS interval (Theorem 1), the vertex property (Lemma 1), validity and finite termination of the Benders certificate (Theorems 2–4), nesting of the AC-corrected sets (Proposition 1) and the conformal no-miss guarantee (Theorem 5). Square brackets name the implementing function in `genco_rss/`. The statements are also checked numerically by `tests/test_theory.py`.

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1 DC Model of the Robust Secure Set

1.1 Sets, data and decision variables

Buses $\mathcal{N}$ ($|\mathcal{N}| = 118$) with reference bus n_0; branches $\mathcal{L}$ ($|\mathcal{L}| = 186$) with reactance x_ℓ and MVA rating r_ℓ (p.u., base 100 MVA); dispatchable units $\mathcal{G}$ ($|\mathcal{G}| = 19$) with the balancing (slack) unit $g_0 \in \mathcal{G}$ located at n_0; PV buses $\mathcal{R}$ ($|\mathcal{R}| = 20$); load buses $\mathcal{D}$; topologies $s \in \mathcal{S} = \{0, 1, 2\}$ (base case; one circuit of 69–70 lost; one circuit of 100–103 lost). Incidence matrices $C_G \in \{0,1\}^{|\mathcal{N}|\times|\mathcal{G}|}$, C_R, C_D map units, PV injections and loads to buses.

Uncertain vector (nature) $p = [P_R; P_D] \in \mathbb{R}^{n_p}$, $n_p = |\mathcal{R}| + |\mathcal{D}|$. Dispatch (operator) $u \in \mathbb{R}^{n_u}_{\geq 0}$, $n_u = |\mathcal{G}|$ (plus $2|\mathcal{R}|$ battery variables in Section 7).

1.2 Power transfer distribution factors

[`grid.ptdf`] Let $C \in \{-1, 0, 1\}^{|\mathcal{L}|\times|\mathcal{N}|}$ be the branch–bus incidence matrix and $X_s = \mathrm{diag}(x^s_\ell)$ with

$$x^s_\ell = \begin{cases} 2x_\ell & \ell \text{ is the corridor of } s \text{ (one of two identical parallel circuits lost)}, \\ x_\ell & \text{otherwise}, \end{cases} \tag{1}$$

since two identical circuits in parallel have half the reactance of one. As in the dissertation data, the corridor rating is kept. With $B^s_f = X_s^{-1}C$, $B^s_{\text{bus}} = C^\top X_s^{-1} C$ and $\mathcal{N}' = \mathcal{N} \setminus \{n_0\}$,

$$H_s[:, \mathcal{N}'] = B^s_f[:, \mathcal{N}'] \big(B^s_{\text{bus}}[\mathcal{N}', \mathcal{N}']\big)^{-1}, \qquad H_s[:, n_0] = 0. \tag{2}$$

The DC branch flows of an operating point in topology s are

$$f_s(p, u) = H_s\big(C_G u + C_R P_R - C_D P_D\big). \tag{3}$$

Because the slack unit sits at n_0, column $H_s C_G[:, g_0] = 0$: the slack unit does not appear in any line constraint (used in Lemma 3).

1.3 Constraints

[`grid.rss_matrices`] For derating factors $\kappa_s \in (0, 1]^{|\mathcal{L}|}$ ($\kappa \equiv 1$ in the DC study, Section 4 otherwise):

$$-\kappa_s \circ r \leq f_s(p, u) \leq \kappa_s \circ r, \qquad s \in \mathcal{S}, \tag{4}$$

$$\mathbf{1}^\top u + \mathbf{1}^\top P_R - \mathbf{1}^\top P_D = 0, \qquad \text{(lossless balance)} \tag{5}$$

$$\underline{u} \leq u \leq \overline{u}. \tag{6}$$

[`grid.gen_limits`, `rss.dc_opf`] With u^0 the linear-cost DC-OPF dispatch at the box centre and ramp fraction $\rho = 0.3$:

$$\underline{u}_k = \max(P^{\min}_k, u^0_k - \rho P^{\max}_k), \quad \overline{u}_k = \min(P^{\max}_k, u^0_k + \rho P^{\max}_k) \ (k \neq g_0), \quad [\underline{u}_{g_0}, \overline{u}_{g_0}] = [P^{\min}_{g_0}, P^{\max}_{g_0}]. \tag{7}$$

Stacking (4)–(6) row-wise gives the polyhedral system

$$Ap + Bu \leq b, \qquad A = \begin{bmatrix} A_\ell \\ -A_\ell \\ a^\top \\ -a^\top \\ 0 \\ 0 \end{bmatrix}, \ B = \begin{bmatrix} B_\ell \\ -B_\ell \\ \mathbf{1}^\top \\ -\mathbf{1}^\top \\ -I \\ I \end{bmatrix}, \ b = \begin{bmatrix} \kappa \circ r \\ \kappa \circ r \\ 0 \\ 0 \\ -\underline{u} \\ \overline{u} \end{bmatrix}, \tag{8}$$

with $A_\ell = [H_s C_R, -H_s C_D]_{s\in\mathcal{S}}$, $B_\ell = [H_s C_G]_{s\in\mathcal{S}}$ (stacked over s), $a = [\mathbf{1}; -\mathbf{1}]$. The follower's feasible set is

$$\mathcal{U}(p) = \{u \in \mathbb{R}^{n_u}_{\geq 0} : Bu \leq b - Ap\}. \tag{9}$$

1.4 Uncertainty set

[`grid.scenario_box`] For hour h, with PV forecast $\hat{P}_R$, capacity $\bar{P}_R$, forecast load $\hat{P}_D$, $e_{\mathrm{PV}}(h) = 0.3\,h/24$ and $e_D(h) = 0.03\,h/24$:

$$\mathcal{P}_h = \Big\{ p :\ \big[\hat{P}_R - e_{\mathrm{PV}}\bar{P}_R\big]_0^{\bar{P}_R} \le P_R \le \big[\hat{P}_R + e_{\mathrm{PV}}\bar{P}_R\big]_0^{\bar{P}_R},\ (1-e_D)\hat{P}_D \le P_D \le (1+e_D)\hat{P}_D \Big\}, \quad (10)$$

where $[\cdot]_0^{\bar{P}}$ clips to $[0, \bar{P}]$ and $P_R \equiv 0$ for $h \notin [7, 18]$. $\mathcal{P}_h = [\underline{p}, \overline{p}]$ is a box with vertex set $V(\mathcal{P}_h)$.

1.5 Robust secure set

The RSS index is the total output of the non-balancing units,

$$\alpha = d^\top u, \qquad d_k = 1\ (k \neq g_0),\ d_{g_0} = 0. \quad (11)$$

Assumption 1. *$\mathcal{U}(p) \neq \emptyset$ for every $p \in \mathcal{P}_h$ (the follower can always serve the load; checked by Benders, which otherwise reports infeasibility).*

Define the follower value functions and the RSS limits

$$\varphi(p) = \max\{d^\top u : u \in \mathcal{U}(p)\}, \qquad \psi(p) = \min\{d^\top u : u \in \mathcal{U}(p)\}, \quad (12)$$

$$\overline{\alpha}_h = \min_{p \in \mathcal{P}_h} \varphi(p), \qquad \underline{\alpha}_h = \max_{p \in \mathcal{P}_h} \psi(p). \quad (13)$$

Theorem 1 (Meaning of the RSS interval)**.** *Under Assumption 1, a value α is achievable for* every *realisation, i.e. $\forall p \in \mathcal{P}_h\ \exists u \in \mathcal{U}(p) : d^\top u = \alpha$, if and only if $\alpha \in [\underline{\alpha}_h, \overline{\alpha}_h]$. In particular the robust secure set is empty if and only if $\underline{\alpha}_h > \overline{\alpha}_h$.*

Proof. Fix p. $\mathcal{U}(p)$ is a non-empty polytope (bounded by (6)), hence compact and convex; its image under the linear map $u \mapsto d^\top u$ is a compact interval, which by (12) equals $[\psi(p), \varphi(p)]$. ($\Leftarrow$) If $\underline{\alpha}_h \le \alpha \le \overline{\alpha}_h$, then for every p, $\psi(p) \le \underline{\alpha}_h \le \alpha \le \overline{\alpha}_h \le \varphi(p)$, so α lies in the image interval and some $u \in \mathcal{U}(p)$ attains it. ($\Rightarrow$) If $\alpha > \overline{\alpha}_h$, take $p^\star \in \arg\min \varphi$ (exists by Lemma 1); then $\alpha > \varphi(p^\star)$ and no $u \in \mathcal{U}(p^\star)$ attains α. The case $\alpha < \underline{\alpha}_h$ is symmetric with $\arg\max \psi$. □

Lemma 1 (Worst cases are vertices)**.** *φ is concave and ψ is convex on $\mathcal{P}_h$. Hence*

$$\overline{\alpha}_h = \min_{p \in V(\mathcal{P}_h)} \varphi(p), \qquad \underline{\alpha}_h = \max_{p \in V(\mathcal{P}_h)} \psi(p).$$

Proof. Let $p_1, p_2 \in \mathcal{P}_h$, $\theta \in [0,1]$, and $u_i \in \mathcal{U}(p_i)$ optimal for $\varphi(p_i)$. Then $u_\theta = \theta u_1 + (1-\theta)u_2 \ge 0$ and $Bu_\theta = \theta B u_1 + (1-\theta)Bu_2 \le b - A(\theta p_1 + (1-\theta)p_2)$, so $u_\theta \in \mathcal{U}(p_\theta)$ and $\varphi(p_\theta) \ge d^\top u_\theta = \theta\varphi(p_1) + (1-\theta)\varphi(p_2)$: φ is concave. The same argument with minimisers shows that ψ is convex. A concave function on a polytope attains its minimum at a vertex (write any point as a convex combination of vertices and use concavity: $\varphi(\sum \theta_j v_j) \ge \sum \theta_j \varphi(v_j) \ge \min_j \varphi(v_j)$); symmetrically a convex function attains its maximum at a vertex. □

2 Bilevel Form, KKT Conditions and the Big-M Baseline

Both limits are instances of the linear bilevel program [`benders.solve`]

$$\mathrm{OPT} = \max_{p \in \mathcal{P}_h,\, u} \ c_1^\top p + d_1^\top u \quad \text{s.t.} \quad u \in \arg\max_{v \in \mathcal{U}(p)} d_2^\top v, \quad (14)$$

with $c_1 = 0$ and $(d_1, d_2) = (-d, d)$ for the upper limit ($\overline{\alpha}_h = -\mathrm{OPT}$) and $(d_1, d_2) = (d, -d)$ for the lower limit ($\underline{\alpha}_h = \mathrm{OPT}$).

Remark 1. Since $d_1 = -d_2$, every follower optimum gives the leader the same value $d_1^\top u = -\max_v d_2^\top v$; optimistic and pessimistic bilevel solutions coincide. For general data we use the optimistic convention: the attained value at p is

$$F(p) = c_1^\top p + \max\{d_1^\top u :\ u \in \mathcal{U}(p),\ d_2^\top u = \max_{v \in \mathcal{U}(p)} d_2^\top v\}, \tag{15}$$

computed by two LPs [`benders.follower_response`].

Follower dual and KKT. The follower LP $\max\{d_2^\top u : Bu \le b - Ap,\ u \ge 0\}$ has the dual $\min\{\lambda^\top(b - Ap) :\ B^\top\lambda \ge d_2,\ \lambda \ge 0\}$. A feasible u is optimal iff there is λ with

$$\lambda \ge 0, \quad B^\top\lambda \ge d_2, \quad u_k\,(B^\top\lambda - d_2)_k = 0\ \forall k, \quad \lambda_i\,(b - Ap - Bu)_i = 0\ \forall i. \tag{16}$$

Let $\mathcal{R} = \{i : B_i \ne 0\}$. Rows with $B_i = 0$ do not enter $B^\top\lambda$; setting $\lambda_i = 0$ there preserves (16), so w.l.o.g. λ is supported on $\mathcal{R}$.

Big-M KKT MILP of the dissertation [`rss.bigm_milp`, `RSS_MILP2.m`]. With $x = (p, u, \lambda)$, $G(x) = \big(0;\ B^\top\lambda - d_2;\ b - Ap - Bu\big)$ and binaries w:

$$\max\ c_1^\top p + d_1^\top u \ \text{ s.t. } \ Ap + Bu \le b, \ \ 0 \le x \le Lw, \ \ 0 \le G(x) \le L(\mathbf{1} - w), \ \ p \in \mathcal{P}_h, \tag{17}$$

$L = 5 \cdot 10^4$. If L bounds every component of some optimal KKT pair, (17) is exact.

Remark 2 (Why the big-M MILP misreports). (i) If L is smaller than the components of every optimal KKT pair, (17) cuts the optimum off [1]. (ii) A MILP solver accepts $w_j \le \varepsilon_I$ as 0 (integrality tolerance, $\varepsilon_I = 10^{-6}$ in HiGHS); then $x_j \le L\varepsilon_I = 0.05$ while $G_j(x) > 0$ is allowed, i.e. complementarity can be violated by up to $L\varepsilon_I$ per pair. The reported u is then not follower-optimal and the reported objective is not attained. The reporting errors measured in E1 (up to 0.063 p.u.) are of this order. Our method never reports a value that is not attained: the incumbent is always $F(p)$ of (15), computed by LP.

3 Two-Phase Combinatorial Benders Decomposition

3.1 Exact bounds

For $i \in \mathcal{R}$ the largest possible slack over the box is

$$S_i = \max\Big(0,\ b_i - \min_{p \in \mathcal{P}_h,\, 0 \le u \le \overline{u}} (A_i p + B_i u)\Big) = \max\Big(0,\ b_i - \textstyle\sum_j \min(A_{ij}\underline{p}_j, A_{ij}\overline{p}_j) - \sum_k \min(0, B_{ik}\overline{u}_k)\Big), \tag{18}$$

and $\overline{U}_k = \overline{u}_k$. Both are computed from the data; no constant is chosen.

3.2 Phase-1 master

For a finite set $\mathcal{C}$ of cuts, the master problem MP($\mathcal{C}$) has variables p, u and binaries $z^u \in \{0,1\}^{n_u}$, $z^\lambda \in \{0,1\}^{|\mathcal{R}|}$:

$$\begin{aligned}
&\max\ c_1^\top p + d_1^\top u && && (19)\\
&\text{s.t. } Ap + Bu \le b, \quad p \in \mathcal{P}_h, \quad u \ge 0, && && (\text{M1})\\
&\quad u_k \le \overline{U}_k\, z^u_k, && k = 1..n_u && (\text{M2})\\
&\quad b_i - A_i p - B_i u \le S_i\,(1 - z^\lambda_i), && i \in \mathcal{R} && (\text{M3})\\
&\quad z^\lambda_a + z^\lambda_j \le 1, && (a, j) \in \Pi && (\text{M4})\\
&\quad d_2^\top u \ge \varphi_R, && \text{if } \varphi_R > -\infty && (\text{M5})\\
&\quad \textstyle\sum_{k \in C_u}(1 - z^u_k) + \sum_{i \in C_\ell} z^\lambda_i \ge 1, && (C_u, C_\ell) \in \mathcal{C} && (\text{M6})
\end{aligned}$$

where Π is the set of opposite row pairs, $[A_a\ B_a] = -[A_j\ B_j]$ with $b_a + b_j > 0$ (e.g. the two sides of a line limit), and φ_R is defined in (22). Interpretation: $z^u_k = 0$ forces $u_k = 0$; $z^\lambda_i = 1$ forces row i to be binding; $z^\lambda_i = 0$ will mean $\lambda_i = 0$.

3.3 Subproblem, Farkas certificate and cut

For index sets $J_u \subseteq \{1..n_u\}$ (units allowed to be positive) and $J_\ell \subseteq \mathcal{R}$ (rows with positive slack) define the dual system

$$\mathrm{DF}(J_u, J_\ell): \quad \lambda \in \mathbb{R}^{\mathcal{R}}_{\geq 0}, \quad \lambda_i = 0 \ (i \in J_\ell), \quad B_{\mathcal{R}}^\top \lambda \geq d_2, \quad (B_{\mathcal{R}}^\top \lambda)_k = d_{2,k} \ (k \in J_u). \tag{20}$$

[`benders._dual_feasible`] Given a master solution $(\hat{p}, \hat{u}, \hat{z})$ with slack $\hat{s} = b - A\hat{p} - B\hat{u}$, the subproblem checks $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$, $J_u(\hat{u}) = \{k : \hat{u}_k > 0\}$, $J_\ell(\hat{s}) = \{i \in \mathcal{R} : \hat{s}_i > 0\}$.

Lemma 2 (Farkas certificate)**.** *Let $\tilde{B} = B_{\mathcal{R}\setminus J_\ell}$ (the rows whose multiplier may be positive).* $\mathrm{DF}(J_u, J_\ell)$ *is infeasible if and only if there is $y \in \mathbb{R}^{n_u}$ with*

$$\tilde{B}y \leq 0, \qquad y_k \geq 0 \ (k \notin J_u), \qquad d_2^\top y > 0. \tag{21}$$

Proof. Eliminating $\lambda_{J_\ell} = 0$ and introducing surplus $t \geq 0$ with $t_{J_u} = 0$, DF reads $\{(\lambda, t_{\bar{J}_u}) \geq 0 : \tilde{B}^\top \lambda - I_{:,\bar{J}_u} t_{\bar{J}_u} = d_2\}$, $\bar{J}_u$ the complement of J_u. By Farkas' lemma, $\{x \geq 0 : Mx = d_2\}$ is infeasible iff some y has $M^\top y \leq 0$ and $d_2^\top y > 0$; with $M = [\tilde{B}^\top, -I_{:,\bar{J}_u}]$ this is (21). □

[`benders._farkas_subset`] A certificate also identifies small infeasible subsystems: an elastic LP relaxes each restriction $\lambda_i = 0$ $(i \in J_\ell)$ and $(B^\top\lambda)_k = d_{2,k}$ $(k \in J_u)$ by a non-negative variable and minimises their sum; the restrictions with a positive dual form $C_\ell \subseteq J_\ell$, $C_u \subseteq J_u$. A second LP checks that $\mathrm{DF}(C_u, C_\ell)$ is infeasible; if not, the full sets $C_u = J_u$, $C_\ell = J_\ell$ are used, which is always correct. Smaller sets give stronger cuts.

If $\mathrm{DF}(C_u, C_\ell)$ is infeasible, the *combinatorial Benders cut* (M6) [2] is added. Equivalently $\sum_{k \in C_u} z^u_k - \sum_{i \in C_\ell} z^\lambda_i \leq |C_u| - 1$.

3.4 Robust value-function cut

[`benders.robust_bound`] Let $F \subseteq \{k : d_{2,k} = 0\}$ be a set of balancing columns such that every row of B either has $B_{iF} = 0$, or is an own bound row of one $f \in F$ (with $A_i = 0$), or belongs to the single equality pair (i_1, i_2) with $[A_{i_1}\ B_{i_1}] = -[A_{i_2}\ B_{i_2}]$ (the balance (5)). For the RSS, $F = \{g_0\}$ (the slack unit has a zero PTDF column, Section 1); if the structure is not met, $F = \emptyset$. Let $[\underline{w}_f, \overline{w}_f]$ be the own-bound interval of $f \in F$, X the other columns, $\ell_F = \sum_f \min(B_{i_1 f}\underline{w}_f, B_{i_1 f}\overline{w}_f)$, $h_F = \sum_f \max(B_{i_1 f}\underline{w}_f, B_{i_1 f}\overline{w}_f)$ and $\overline{A_i p} = \max_{p \in \mathcal{P}_h} A_i p = \sum_j \max(A_{ij}\underline{p}_j, A_{ij}\overline{p}_j)$. Define the *static robust LP*

$$\begin{aligned} \varphi_R = \max_{0 \leq v \leq \overline{u}_X} \ d_{2,X}^\top v \quad \text{s.t.} \quad & B_{iX} v \leq b_i - \overline{A_i p} \qquad (i \notin \{i_1, i_2\},\ B_{iF} = 0), \\ & B_{i_1 X} v \leq b_{i_1} - \overline{A_{i_1} p} - \ell_F, \qquad B_{i_2 X} v \leq b_{i_2} - \overline{A_{i_2} p} + h_F. \end{aligned} \tag{22}$$

Lemma 3 (Robust value-function bound)**.** *For every $p \in \mathcal{P}_h$:* $\max\{d_2^\top u : u \in \mathcal{U}(p)\} \geq \varphi_R$.

Proof. If (22) is infeasible, $\varphi_R = -\infty$ and there is nothing to prove. Let v be optimal in (22) and fix p. Rows without F: $A_i p + B_{iX} v \leq \overline{A_i p} + B_{iX} v \leq b_i$. Equality pair: write $g = A_{i_1} p + B_{i_1 X} v$, so the pair reads $-b_{i_2} \leq g + B_{i_1 F} w \leq b_{i_1}$. As w ranges over the box $\prod_f [\underline{w}_f, \overline{w}_f]$, $B_{i_1 F} w$ covers the whole interval $[\ell_F, h_F]$ (a linear image of a box is an interval). The constraints of (22) give $g \leq b_{i_1} - \ell_F$ (row i_1) and $-g \leq b_{i_2} + h_F$ (row i_2, since $A_{i_2} = -A_{i_1}$, $B_{i_2 X} = -B_{i_1 X}$), i.e. $[-b_{i_2} - g,\ b_{i_1} - g] \cap [\ell_F, h_F] \neq \emptyset$ because $b_{i_1} + b_{i_2} \geq 0$. Pick $w(p)$ in the box with $B_{i_1 F} w(p)$ in this intersection; own-bound rows of F hold by construction. Hence $u = (v, w(p)) \in \mathcal{U}(p)$ and, as $d_{2,F} = 0$, $\max_{u' \in \mathcal{U}(p)} d_2^\top u' \geq d_{2,X}^\top v = \varphi_R$. □

The dual multipliers μ of (22) give the vertex $p_j^{\mathrm{adv}} = \overline{p}_j$ if $(A^\top \mu)_j > 0$, else $\underline{p}_j$, which maximises $\mu^\top A p$ over the box; $F(p^{\mathrm{adv}})$ is the first incumbent (a heuristic; any feasible p is valid).

3.5 Validity of the master bound

Theorem 2 (Master relaxation)**.** *Let (p, u^*) be bilevel feasible, i.e. $p \in \mathcal{P}_h$ and u^* optimal for the follower at p, and let all cuts in $\mathcal{C}$ be generated as above. Then there is z such that (p, u^*, z) is feasible for* $\mathrm{MP}(\mathcal{C})$*. Consequently* $\mathrm{UB}(\mathcal{C}) := \max \mathrm{MP}(\mathcal{C}) \geq \mathrm{OPT}$.

Proof. Let λ^* satisfy (16) with u^*, supported on $\mathcal{R}$. Set $z^u_k = \mathbb{1}[u^*_k > 0]$ and $z^\lambda_i = \mathbb{1}[\lambda^*_i > 0]$. (M1) holds by feasibility. (M2): if $u^*_k > 0$ then $z^u_k = 1$ and $u^*_k \leq \overline{u}_k$ by (6); otherwise $u^*_k = 0$. (M3): if $\lambda^*_i > 0$, complementarity gives zero slack, which is the requirement for $z^\lambda_i = 1$; if $z^\lambda_i = 0$ the slack is at most S_i by (18). (M4): if $\lambda^*_a, \lambda^*_j > 0$ both rows are binding, $[A_a\, B_a](p; u^*) = b_a$ and $-[A_a\, B_a](p; u^*) = b_j$, so $b_a + b_j = 0$, contradicting $(a, j) \in \Pi$. (M5): $d_2^\top u^* = \max_{\mathcal{U}(p)} d_2^\top u \geq \varphi_R$ by Lemma 3. (M6): suppose a cut (C_u, C_ℓ) is violated, i.e. $z^u_k = 1$ for all $k \in C_u$ and $z^\lambda_i = 0$ for all $i \in C_\ell$. Then $u^*_k > 0$ on C_u, so $(B^\top \lambda^*)_k = d_{2,k}$ there by complementarity, and $\lambda^*_i = 0$ on C_ℓ; together with $\lambda^* \geq 0$, $B^\top \lambda^* \geq d_2$ this means λ^* solves $\mathrm{DF}(C_u, C_\ell)$, contradicting how the cut was generated. Hence (p, u^*, z) is master-feasible and its objective, the leader value of (p, u^*), is at most $\mathrm{UB}(\mathcal{C})$. Taking (p, u^*) optimal for (14) gives the claim. □

Lemma 4 (Attained incumbent)**.** *For every $p \in \mathcal{P}_h$, $F(p) \leq \mathrm{OPT}$ and $F(p)$ is attained by the bilevel feasible point $(p, u(p))$ returned by* `follower_response`*.*

Proof. $u(p)$ maximises $d_2^\top u$ over $\mathcal{U}(p)$ and, among those maximisers, $d_1^\top u$; so $(p, u(p))$ is feasible for (14) and its value $F(p)$ is at most the optimum. □

Theorem 3 (Certificate and finite termination of phase 1)**.** *Let* $\mathrm{LB}_t = \max_{\tau \leq t} F(\hat{p}_\tau)$ *(incumbents, also local search) and* $\mathrm{UB}_t = \min_{\tau \leq t} \mathrm{UB}(\mathcal{C}_\tau)$*. Then for every iteration t,*

$$\mathrm{LB}_t \leq \mathrm{OPT} \leq \mathrm{UB}_t, \tag{23}$$

so $(\mathrm{UB}_t - \mathrm{LB}_t)/\max(1, |\mathrm{UB}_t|)$ *is a certified relative gap. In exact arithmetic and without the lift to phase 2, phase 1 terminates after at most $2^{n_u + |\mathcal{R}|}$ iterations, either (a) because* $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$ *is feasible, in which case $(\hat{p}, \hat{u})$ is optimal and* $\mathrm{LB} = \mathrm{UB} = \mathrm{OPT}$*, or (b) because the gap is below the tolerance.*

Proof. (23) follows from Lemma 4 and Theorem 2 (the cut set only grows, so every $\mathrm{UB}(\mathcal{C}_\tau)$ is a valid bound).

(a) If $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$ has a solution λ, then $\lambda \geq 0$, $B^\top \lambda \geq d_2$, $\lambda_i = 0$ wherever $\hat{s}_i > 0$ and $(B^\top \lambda - d_2)_k = 0$ wherever $\hat{u}_k > 0$: (16) holds, so $\hat{u}$ is follower-optimal at $\hat{p}$ and $(\hat{p}, \hat{u})$ is bilevel feasible with value $\mathrm{UB}(\mathcal{C}_t)$. With (23), $\mathrm{UB}(\mathcal{C}_t) = \mathrm{OPT}$.

Otherwise a cut is added that is violated by the current binary vector $\hat{z}$: consider $J^z_u = \{k : \hat{z}^u_k = 1\} \supseteq J_u(\hat{u})$ (by (M2)) and $J^z_\ell = \{i : \hat{z}^\lambda_i = 0\} \supseteq J_\ell(\hat{s})$ (by (M3)). Restricting a system by fewer equalities/zeros is a relaxation, so $\mathrm{DF}(J^z_u, J^z_\ell)$ feasible would imply $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$ feasible; hence $\mathrm{DF}(J^z_u, J^z_\ell)$ is infeasible. The implementation uses the Farkas subsets of $(J_u(\hat{u}), J_\ell(\hat{s}))$ if $\hat{z}$ violates the resulting cut and otherwise those of (J^z_u, J^z_ℓ); in both cases $C_u \subseteq J^z_u$ and $C_\ell \subseteq J^z_\ell$, so the left-hand side of (M6) is 0 at $\hat{z}$: the cut removes $\hat{z}$. By Theorem 2 it never removes the binary vector of an optimal KKT pair, so the master stays feasible. Each iteration removes at least one of the finitely many binary vectors, which proves finite termination. □

Remark 3. UB is taken from the master solution with relative MIP gap 10^{-8}; the bound is therefore valid up to that tolerance. Tolerances $\hat{u}_k > 10^{-5}\overline{u}_k$ and $\hat{s}_i > 10^{-5} S_i$ define J_u, J_ℓ numerically.

3.6 Phase 2: dual lifting

If (22) is infeasible or phase 1 exceeds 15 iterations, the master is lifted with dual variables $\lambda \in \mathbb{R}^{\mathcal{R}}_{\geq 0}$ and the constraints

$$B^\top_{\mathcal{R}} \lambda \geq d_2, \qquad B^\top_{\mathcal{R}} \lambda - d_2 \leq L_d(\mathbf{1} - z^u), \qquad \lambda \leq L_d\, z^\lambda, \tag{24}$$

($L_d = 5 \cdot 10^4$; a big-M on the dual side only), keeping (M1)–(M6).

Lemma 5 (A uniform dual bound exists)**.** *The dual feasible set* $\mathcal{D} = \{\lambda \in \mathbb{R}^{\mathcal{R}}_{\geq 0} : B^{\top}_{\mathcal{R}}\lambda \geq d_2\}$ *does not depend on* p*. For every* p *there is an optimal dual solution at a vertex of* $\mathcal{D}$*. Hence* $L^* = \max_{\lambda \in \mathrm{vert}(\mathcal{D})} \max(\|\lambda\|_\infty, \|B^{\top}\lambda - d_2\|_\infty) < \infty$ *bounds an optimal KKT multiplier for every* p *simultaneously.*

Proof. $\mathcal{D}$ involves only B and d_2. It lies in the non-negative orthant, so it is a pointed polyhedron with finitely many vertices. Under Assumption 1 the follower LP is feasible and bounded (by (6)), so by strong duality the dual $\min_{\lambda \in \mathcal{D}} \lambda^{\top}(b - Ap)$ has a finite optimum, attained at a vertex (fundamental theorem of LP). □

Theorem 4 (Phase-2 exactness)**.** *Every feasible point of the lifted master is bilevel feasible. If* $L_d \geq L^*$*, the lifted master optimum equals* OPT*.*

Proof. At a lifted-master point: $z^u_k = 0 \Rightarrow u_k = 0$ by (M2); $z^u_k = 1 \Rightarrow (B^{\top}\lambda - d_2)_k = 0$ by (24); so $u_k(B^{\top}\lambda - d_2)_k = 0$. Likewise $z^{\lambda}_i = 0 \Rightarrow \lambda_i = 0$ and $z^{\lambda}_i = 1 \Rightarrow$ zero slack by (M3). Hence (16) holds and u is follower-optimal: the point is bilevel feasible and its value is $\leq$ OPT. Conversely take an optimal (p, u^*) and a vertex multiplier λ^* (Lemma 5); with z as in the proof of Theorem 2, (M1)–(M6) hold and (24) holds because $\lambda^*_i \leq L^* \leq L_d$ and $(B^{\top}\lambda^* - d_2)_k \leq L^* \leq L_d$ (and $= 0$ when $z^u_k = 1$). So the lifted master value is $\geq$ OPT. □

Phase 2 was needed in 1 of 48 problems (hour 21, upper); its certificate is conditional on $L_d \geq L^*$, whereas phase 1 is unconditional.

3.7 Algorithm

1. Compute S (18), Π, φ_R and p^{adv} (22); LB $\leftarrow F(p^{\mathrm{adv}})$, $\mathcal{C} \leftarrow \emptyset$.
2. Solve MP($\mathcal{C}$) (lifted if phase 2) $\rightarrow (\hat{p}, \hat{u}, \hat{z})$, UB $\leftarrow \min(\mathrm{UB}, \mathrm{val})$.
3. LB $\leftarrow \max(\mathrm{LB}, F(\hat{p}))$; in iteration 1, one-flip local search over vertices.
4. Stop if phase 2, if the gap $\leq 10^{-3}$, or if $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$ is feasible.
5. Otherwise add the Farkas cut (M6) and go to 2; lift (phase 2) after 15 iterations.

4 AC Model, Security and the AC-Correction Loop

4.1 AC power flow

[gridfm-datakit + lightsim2grid] For an operating point (h, s, p, u) the bus voltages $V_i\angle\theta_i$ solve, with the topology-s admittance matrix $Y^s = G^s + jB^s$,

$$P^{\mathrm{inj}}_i = V_i \textstyle\sum_j V_j\big(G^s_{ij}\cos\theta_{ij} + B^s_{ij}\sin\theta_{ij}\big), \tag{25}$$

$$Q^{\mathrm{inj}}_i = V_i \textstyle\sum_j V_j\big(G^s_{ij}\sin\theta_{ij} - B^s_{ij}\cos\theta_{ij}\big), \tag{26}$$

where $P^{\mathrm{inj}} = C_G u + C_R P_R - C_D P_D$ at PQ and PV buses (slack injection free), $Q^{\mathrm{inj}} = -C_D(\gamma \circ P_D)$ at PQ buses with the fixed load power factor $\gamma_i = Q^0_{D,i}/P^0_{D,i}$, generator buses at the AC-OPF set-point $V_i = V^{\mathrm{set}}_i(h)$, and $\theta_{n_0} = 0$. The contingency doubles the series impedance $r_\ell + jx_\ell$ and halves the charging susceptance of the corridor (one of two identical circuits lost), consistent with (1); PV injects at unity power factor. Branch $\ell = (i, j)$ in the π-model carries $S^{\mathrm{f}}_\ell = V_i\angle\theta_i\,(I^{\mathrm{f}}_\ell)^*$, $S^{\mathrm{t}}_\ell = V_j\angle\theta_j\,(I^{\mathrm{t}}_\ell)^*$ and $S_\ell = \max(|S^{\mathrm{f}}_\ell|, |S^{\mathrm{t}}_\ell|)$.

4.2 Security quantities

[`acrss.read_ac`, `genco.evaluate`] With PQ buses $\mathcal{N}_{PQ}$, limits $[\underline{V}_i, \overline{V}_i]$ and tolerance $\tau_V = 0.01$ p.u.,

$$q_{\text{load}} = \max_{\ell} \frac{S_\ell}{\overline{S}_\ell}, \qquad q_{\text{vlow}} = \max_{i \in \mathcal{N}_{PQ}} (\underline{V}_i - V_i), \qquad q_{\text{vhigh}} = \max_{i \in \mathcal{N}_{PQ}} (V_i - \overline{V}_i), \tag{27}$$

and the scenario is *secure* iff

$$q_{\text{load}} \le 1, \qquad q_{\text{vlow}} \le \tau_V, \qquad q_{\text{vhigh}} \le \tau_V. \tag{28}$$

For verified scenarios the slack output is also checked, $P_{g_0} \le P_{g_0}^{\max}$.

4.3 AC-correction update

[`acrss.ac_rss`] At iteration t the DC model uses κ^t and $\overline{u}_{g_0}^t$. For each verified overload ($q_{\text{load},\ell} = S_\ell/\overline{S}_\ell > 1$ in topology s), with $\varepsilon = 0.01$ and floor $\kappa_{\min} = 0.8$,

$$\kappa_{s\ell}^{t+1} = \max\Big(\kappa_{\min},\ \min\Big(\kappa_{s\ell}^t,\ \kappa_{s\ell}^t \frac{1-\varepsilon}{S_\ell/\overline{S}_\ell}\Big)\Big), \qquad \overline{u}_{g_0}^{t+1} = \overline{u}_{g_0}^t - (1+\varepsilon) \max_{\text{verified}} (P_{g_0} - P_{g_0}^{\max})^+. \tag{29}$$

Proposition 1 (Nesting)**.** *If $\kappa' \le \kappa$ componentwise and $\overline{u}' \le \overline{u}$, then $\mathcal{U}'(p) \subseteq \mathcal{U}(p)$ for all p, $\varphi' \le \varphi$, $\psi' \ge \psi$, and $[\underline{\alpha}'_h, \overline{\alpha}'_h] \subseteq [\underline{\alpha}_h, \overline{\alpha}_h]$. In particular the iterates of (29) produce nested intervals, all inside the DC interval.*

Proof. Only right-hand sides change: the line rows (4) and the bound $\overline{u}_{g_0}$ become tighter, so every $u \in \mathcal{U}'(p)$ satisfies the constraints of $\mathcal{U}(p)$. A maximum over a subset is not larger and a minimum over a subset is not smaller, hence $\varphi' \le \varphi$, $\psi' \ge \psi$, and by (13) $\overline{\alpha}' \le \overline{\alpha}$, $\underline{\alpha}' \ge \underline{\alpha}$. The update (29) is non-increasing in both κ and $\overline{u}_{g_0}$. □

Corollary 1 (Emptiness is robust to merging the cuts)**.** *The two limits are corrected with their own factors $\kappa^{\text{up}}, \kappa^{\text{lo}}$. If $\underline{\alpha}_h(\kappa^{\text{lo}}) > \overline{\alpha}_h(\kappa^{\text{up}})$, then with the merged factors $\kappa = \min(\kappa^{\text{up}}, \kappa^{\text{lo}})$ the robust secure set is also empty: $\underline{\alpha}_h(\kappa) \ge \underline{\alpha}_h(\kappa^{\text{lo}}) > \overline{\alpha}_h(\kappa^{\text{up}}) \ge \overline{\alpha}_h(\kappa)$ by Proposition 1, and Theorem 1 applies.*

Proposition 2 (Termination and what is certified)**.** *Each $\kappa_{s\ell}$ can strictly decrease at most $\lceil \ln \kappa_{\min} / \ln(1-\varepsilon) \rceil = 23$ times before it reaches the floor, and the loop is capped at 6 iterations. If it stops at iteration t because no verified scenario violates, then: (i) α^t is the certified optimum of the corrected DC bilevel (Theorem 3); (ii) every flagged pool scenario is AC-secure, verified by exact power flow; (iii) every cleared pool scenario that is exchangeable with the calibration data is insecure with probability at most α (Theorem 5), and the expected number of insecure cleared scenarios in a pool of N is at most αN; (iv) with the audit enabled (all our experiments) every pool scenario is solved exactly, and a zero audit count means the whole final pool is thermally secure.*

Proof. A strict decrease multiplies $\kappa_{s\ell}$ by $(1-\varepsilon)/q < 1-\varepsilon$ since $q > 1$; after n decreases $\kappa_{s\ell} \le (1-\varepsilon)^n$, which is below $\kappa_{\min}$ for $n \ge \ln \kappa_{\min} / \ln(1-\varepsilon)$. (i) is Theorem 3 applied to the data of iteration t; (ii) is the stopping rule; (iii) is Theorem 5 plus linearity of expectation; (iv) is the definition of the audit. □

Remark 4. (29) is a correction heuristic, not an exact AC robust counterpart: it tightens the *active-power* limit of a branch whose *apparent* power is too high. If the overload is mostly reactive, the floor $\kappa_{\min}$ is reached (it was reached in every lower-limit problem of E4), and the corrected limits are conservative.

5 GENCO Trajectory, Recalibration and Conformal Screening

5.1 Corrective-layer trajectory

GENCO (`GNS_heterogeneous`) refines a state estimate $\hat{x}^{(l)} = (\hat{V}^{(l)}, \hat{\theta}^{(l)})$ over K layers. In layer l it evaluates the AC mismatch of (25)–(26), $\Delta P_i^{(l)} = P_i^{\text{inj}} - P_i(\hat{x}^{(l)})$, $\Delta Q_i^{(l)} = Q_i^{\text{inj}} - Q_i(\hat{x}^{(l)})$, and feeds it back into the bus embeddings. A forward hook on its residual layer records

$$r_i^{(l)} = \sqrt{(\Delta P_i^{(l)})^2 + (\Delta Q_i^{(l)})^2}, \qquad \bar{r}^{(l)} = \frac{1}{|\mathcal{N}|}\sum_i r_i^{(l)}, \qquad \rho = \bar{r}^{(K)}/\bar{r}^{(1)}. \tag{30}$$

GENCO's predictions $\hat{q}_k$ of (27) use its final state and graphkit's branch-flow operator.

5.2 Physics-trajectory recalibration (PTR)

Let $z = (\hat{q}_{\text{load}}, \hat{q}_{\text{vlow}}, \hat{q}_{\text{vhigh}}, \hat{q}_{\text{qex}}, \hat{q}_{\text{slack}}, \bar{r}^{(1)}, \ldots, \bar{r}^{(K)}, \rho)$ be GENCO-internal features (no hour, no topology label). On a *tune* set $\mathcal{T}$ solved exactly, fit for each k a gradient-boosted regression ensemble

$$g_k = \arg\min_{g \in \mathcal{F}} \sum_{j \in \mathcal{T}} \big(q_{k,j} - \hat{q}_{k,j} - g(z_j)\big)^2 \quad \text{(300 depth-3 trees, rate 0.05, subsample 0.8)}, \tag{31}$$

and set $\tilde{q}_k = \hat{q}_k + g_k(z)$. Plain conformal uses $g_k \equiv 0$; the residual-normalised variant uses $g_k \equiv 0$ with a scale $\sigma_k + \beta_k \bar{r}^{(K)}$ tuned on $\mathcal{T}$.

5.3 Split-conformal screening

Let $\ell_{\text{load}} = 1$, $\ell_{\text{vlow}} = \ell_{\text{vhigh}} = \tau_V$, a level $\alpha \in (0,1)$, $a_k = \alpha/3$, and a calibration set $\mathcal{C}$ of n scenarios solved exactly, independent of $\mathcal{T}$. With scale $\sigma_k(z) > 0$ ($\equiv 1$ for plain and PTR),

$$s_{k,j} = \frac{q_{k,j} - \tilde{q}_{k,j}}{\sigma_k(z_j)}, \qquad c_k = \text{the } \big\lceil (n+1)(1-a_k) \big\rceil\text{-th smallest of } \{s_{k,j}\}_{j \in \mathcal{C}} \ (+\infty \text{ if } > n), \tag{32}$$

and a new scenario is *flagged* (sent to exact AC) iff

$$\exists k: \ \tilde{q}_k + c_k\, \sigma_k(z) > \ell_k; \tag{33}$$

otherwise it is *cleared*.

Lemma 6 (Split-conformal quantile)**.** *Let $S_1, \ldots, S_{n+1}$ be exchangeable real random variables, $k = \lceil (n+1)(1-a) \rceil$ and c the k-th smallest of $S_1, \ldots, S_n$ ($c = +\infty$ if $k > n$). Then* $\Pr[S_{n+1} > c] \le a$.

Proof. If $k > n$ the event is empty. Otherwise, if $S_{n+1} > c$ then at least k of $S_1, \ldots, S_n$ are strictly smaller than S_{n+1}, i.e. $R_{n+1} \ge k$ with $R_i = \#\{j \le n+1 : S_j < S_i\}$. Let $v_{(1)} \le \cdots \le v_{(n+1)}$ be the ordered values. If $R_i \ge k$ then $S_i > v_{(k)}$, which is possible only for indices whose value sits in positions $k+1, \ldots, n+1$; hence $\sum_{i=1}^{n+1} \mathbb{1}[R_i \ge k] \le n+1-k$ surely. By exchangeability $\Pr[R_i \ge k]$ is the same for all i, so $(n+1)\Pr[R_{n+1} \ge k] = \mathbb{E}\sum_i \mathbb{1}[R_i \ge k] \le n+1-k$ and $\Pr[S_{n+1} > c] \le \Pr[R_{n+1} \ge k] \le \frac{n+1-k}{n+1} \le a$, because $k \ge (n+1)(1-a)$. □

Theorem 5 (No-miss guarantee)**.** *Let the calibration scenarios and a test scenario be exchangeable (e.g. i.i.d. draws of the screening distribution) and let g_k, σ_k be fixed given $\mathcal{T}$ (fitted without $\mathcal{C}$ and the test scenario). Then*

$$\Pr[\textit{test scenario insecure by (28) and cleared}] \le \alpha. \tag{34}$$

Proof. Condition on $\mathcal{T}$. Then g_k and σ_k are fixed functions, so the scores $s_{k,1}, \ldots, s_{k,n}, s_{k,n+1}$ (the last for the test scenario) are exchangeable for each k. If the test scenario is insecure, $q_k > \ell_k$ for some k; if it is cleared, $\tilde{q}_k + c_k\sigma_k \le \ell_k < q_k$, i.e. $s_{k,n+1} > c_k$. Hence

$$\{\text{insecure and cleared}\} \subseteq \bigcup_k \{s_{k,n+1} > c_k\}, \qquad \Pr[\cdot \mid \mathcal{T}] \le \sum_k \Pr[s_{k,n+1} > c_k \mid \mathcal{T}] \le \sum_k a_k = \alpha$$

by the union bound and Lemma 6. Taking the expectation over $\mathcal{T}$ gives (34). □

Corollary 2. *The guarantee holds for* every *recalibrator g_k (including $g_k \equiv 0$ and all ablation variants), because the proof uses only that g_k is fixed before calibration. A better g_k cannot weaken it; it only reduces the flag rate. For N exchangeable scenarios the expected number of insecure-and-cleared ones is at most αN.*

Remark 5. The guarantee is marginal (over the screening distribution), not conditional on the hour or topology. For the unseen topology the exchangeability assumption does not hold; the measured miss rates there are an empirical robustness check.

6 GENCO-Gradient Adversary

[`genco.adversarial`] With the dispatch fixed, let $\hat{\ell}_e(p)$, $e \in \mathcal{E}$, be the GENCO-predicted loadings $S_\ell/\overline{S}_\ell$ of all branches in the three topologies (the reactive load moves with P_D at fixed power factor). The objective is the smooth maximum

$$J(p) = \tau \log \sum_{e \in \mathcal{E}} \exp\big(\hat{\ell}_e(p)/\tau\big), \qquad \tau = 0.02. \tag{35}$$

Lemma 7 (Smooth maximum)**.** $\max_e \hat{\ell}_e \le J \le \max_e \hat{\ell}_e + \tau \log|\mathcal{E}|$*, and* $\nabla_p J = \sum_e \pi_e \nabla_p \hat{\ell}_e$ *with* $\pi = \mathrm{softmax}(\hat{\ell}/\tau)$ *in the simplex.*

Proof. With $m = \max_e \hat{\ell}_e$: $e^{m/\tau} \le \sum_e e^{\hat{\ell}_e/\tau} \le |\mathcal{E}| e^{m/\tau}$; take $\tau \log$. The gradient follows from the chain rule, $\partial J/\partial \hat{\ell}_e = e^{\hat{\ell}_e/\tau}/\sum_{e'} e^{\hat{\ell}_{e'}/\tau} = \pi_e$. □

The update is projected sign-gradient ascent on the box,

$$p \leftarrow \Pi_{\mathcal{P}_h}\big(p + \eta\,(\overline{p} - \underline{p}) \circ \operatorname{sign} \nabla_p J(p)\big), \qquad \Pi_{\mathcal{P}_h}(y) = \min(\max(y, \underline{p}), \overline{p}),\ \eta = 0.25, \tag{36}$$

for 6 steps from the riskiest pool points; the result is rounded to the nearest vertex, $p_j \leftarrow \overline{p}_j$ if $p_j > \frac{1}{2}(\underline{p}_j + \overline{p}_j)$ else $\underline{p}_j$ (by Lemma 1 the DC worst case is a vertex), the follower response $u(p)$ is recomputed and the point is verified by exact AC. $\Pi_{\mathcal{P}_h}$ is the Euclidean projection onto a box (separable). The adversary is a heuristic; nothing is certified from it without exact verification.

7 Second-Life EV Batteries

[`bess.bess_power`, `rss._add_bess`] At each PV bus $j \in \mathcal{R}$ a battery of power $P_j^{\mathrm{b}} = \mathrm{SoH}_j \cdot 0.35 \cdot \bar{P}_{R,j}$ adds discharge and charge variables $0 \le u_j^{\mathrm{dis}}, u_j^{\mathrm{ch}} \le P_j^{\mathrm{b}}$ to the follower. Its net injection $u^{\mathrm{dis}} - u^{\mathrm{ch}}$ enters (3) through C_R and the balance (5); it does not enter the index, $d_{\mathrm{dis}} = d_{\mathrm{ch}} = 0$.

Proposition 3 (Batteries can only widen the RSS)**.** $\overline{\alpha}_h^{\mathrm{b}} \ge \overline{\alpha}_h$ *and* $\underline{\alpha}_h^{\mathrm{b}} \le \underline{\alpha}_h$.

Proof. For every $u \in \mathcal{U}(p)$, $(u, 0, 0)$ is feasible with batteries (zero battery variables leave all rows unchanged), with the same index. Hence the index range at each p can only grow, $\varphi^{\mathrm{b}} \ge \varphi$ and $\psi^{\mathrm{b}} \le \psi$, and (13) gives the claim. □

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